



TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2014
Higher 2

CANDIDATE
NAME

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CENTRE
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INDEX
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BIOLOGY

9648/03

Applications Paper and Planning Question

Wednesday, 17 September 2014

Paper 3

2 hours

Additional Materials:

Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show any working or if you do not use appropriate units.

At the end of the examination, fasten the answer papers securely and hand up separately.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's use	
1	/14
2	/14
3	/12
4	/12
5	/20
TOTAL	/72

Answer **all** questions.

For
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- 1 Fig. 1.1 illustrates a form of molecular cloning known as Sequence- and Ligation-Independent Cloning (SLIC). SLIC allows efficient and reproducible assembly of recombinant DNA with multiple gene inserts.

SLIC gene inserts can be generated using polymerase chain reaction (PCR) (Step 1).

The SLIC procedure uses T4 DNA polymerase, which is also a 3' to 5' exonuclease, to generate single-stranded DNA overhangs in the insert and in the linearized vector fragment (Step 2). The resultant DNA overhangs can be about 16 to 18 nucleotides long and rich in G-C.

The gene insert and the linearized vector are then mixed together to form a recombination intermediate (Step 3).

Transformation is then carried out using competent *E. coli*, without the need for addition of DNA ligase (Step 4). Ligation is completed within the *E. coli*.

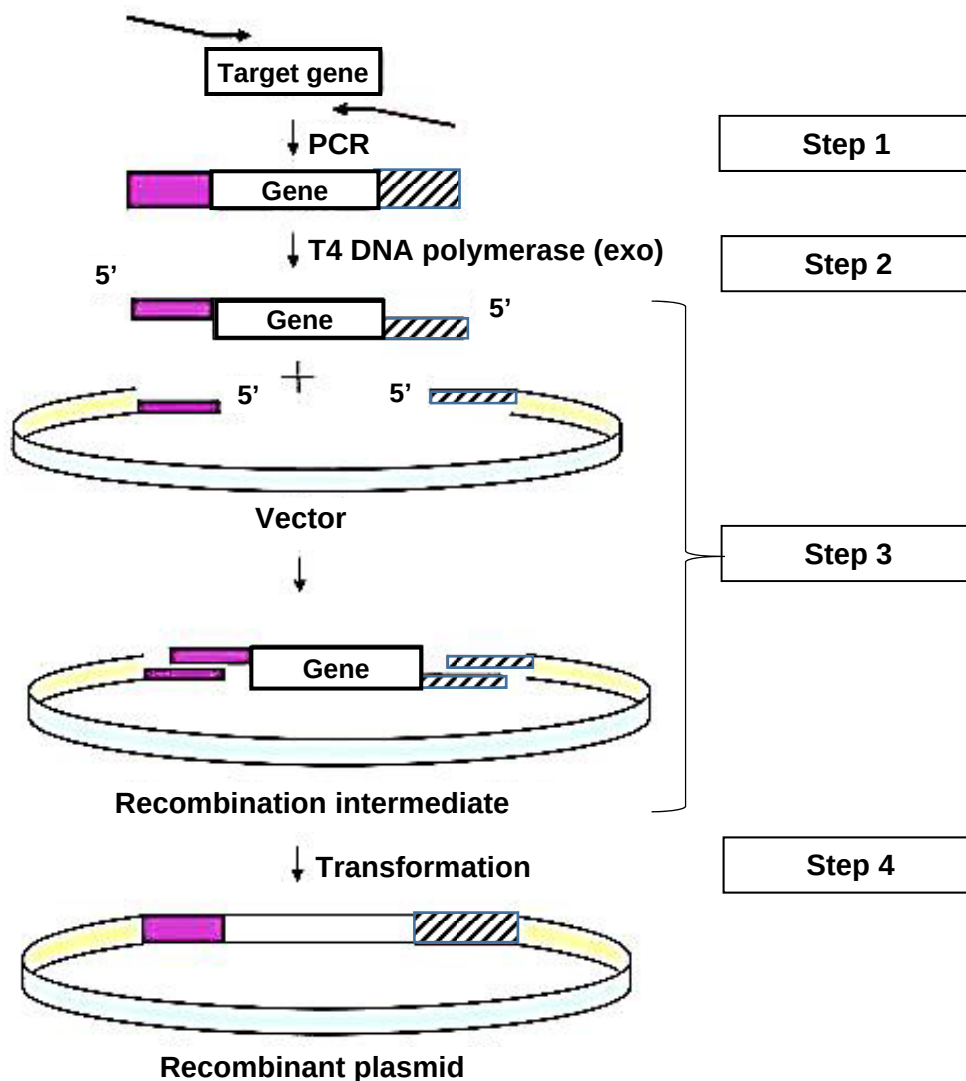


Fig. 1.1

- (a) Describe how the target gene can be obtained from the organism.

[2]

- (b) With reference to **Step 3**,

- (i) explain the advantage of SLIC compared to cloning using restriction endonucleases;

[1]

- (ii) suggest why the addition of DNA ligase is **not** required prior to transformation in SLIC.

[1]

- (c) Explain what is meant by “competent *E. coli*”.

[1]

- (d) Outline the additional steps required if the objective of cloning is to harvest the protein products.

[3]

- (e) Using a named example, explain the advantages of using bacteria to clone human genes.

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[3]

Recently, it has been shown that *Agrobacterium tumefaciens* can transfer recombinant DNA into cultured human cells. The technique used is called protoplast fusion, where the cells are induced to fuse using polyethylene glycol (PEG). Fig. 1.2 shows the cells prior to fusion.

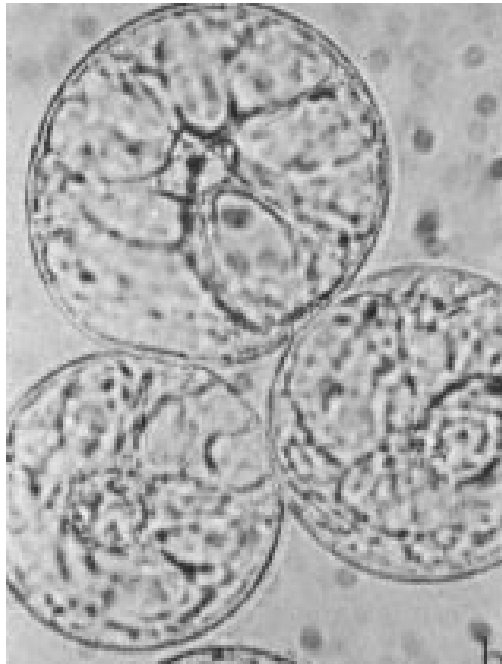


Fig. 1.2

- (f) State how the *Agrobacterium* cells will need to be treated before protoplast fusion.

[1]

- (g) Outline one other method that can be used to transfer recombinant DNA into cultured human cells.

[2]

[Total: 14]

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[Turn over]

- 2 Rice seeds lack the enzymes for two steps in the pathway for β -carotene production. The genes for these two enzymes have been inserted into rice embryos by genetic engineering, giving rise to Golden Rice™. This rice produces β -carotene in the endosperm of the seeds.

(a) Explain why there is a need to produce genetically modified rice which contains β -carotene.

[2]

Fig. 2.1 shows the metabolic pathway by which β -carotene is synthesised in plants, and the enzymes that catalyse each step of the pathway.

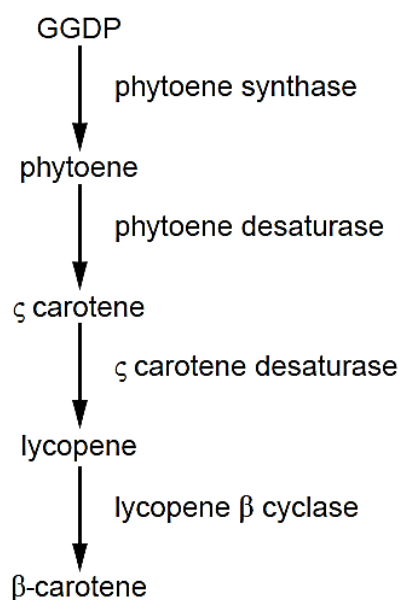


Fig. 2.1

The first types of Golden Rice™ contained a phytoene synthase gene, *psy*, from daffodils and a *crt I* gene, from the bacterium *Erwinia uredovora* which codes for the two desaturase enzymes.

Scientists noticed that these plants produced a very low mass of β -carotene per gram of rice. Measurements of the quantities of intermediates in the metabolic pathway in rice endosperm showed that there was always a large amount of GGDP present but no phytoene accumulated in the tissues.

(b) Explain how these results suggest it was **not** the enzymes produced by the *crt I* gene that was limiting the production of β -carotene.

[1]

[Turn over]

Investigations were carried out to determine if *psy* genes taken from species other than daffodils would enable rice endosperm to produce greater quantities of β -carotene.

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- *Psy* cDNA genes were obtained from maize, tomatoes, peppers and daffodils. The cDNA genes were inserted into different plasmids.
- Glu, a rice endosperm-specific glutelin promoter, and *crt I* genes from *E. uredovora*, were also inserted into all of the plasmids.
- The four types of genetically modified plasmids were then inserted into different cultures of rice cells.
- The quantity of β -carotene produced by these rice cells was measured.

The results are shown in Table 2.1.

Table 2.1

Source of <i>psy</i> cDNA gene	Total β -carotene of rice cells / arbitrary units
Maize	14
Pepper	4
Tomato	6
Daffodil	1

(c) Explain how the *psy* cDNA gene was produced from each of the plant species.

[2]

(d) Suggest why the Glu promoter was inserted into the plasmids.

[1]

[Turn over

- (e) Explain whether the results in Table 2.1 support the hypothesis that the *psy* gene was limiting the production of β -carotene in genetically modified rice.

[2]

- (f) The first type of Golden Rice™ was grown with the *psy* gene from daffodils because daffodils produce large amounts of β -carotene in their yellow petals, and they are monocotyledonous plants, like rice.

Suggest an explanation for the much lower production of β -carotene in rice containing the *psy* gene from daffodils than in rice containing the *psy* gene from maize.

[2]

The following steps were taken to produce Golden Rice™:

- **Step 1 -** A length of cDNA transgenes was made which includes a specific promoter + *psy* gene + *crt* I gene.
- **Step 2 -** Copies of this length of cDNA were inserted into Ti plasmids from the bacterium *Agrobacterium tumefaciens*.
- **Step 3 -** *A. tumefaciens* containing the Ti plasmids were mixed with rice embryos in tissue culture. The T-DNA with the transgenes were inserted into the nuclear DNA of rice embryos.
- **Step 4 -** The embryos were grown into calli, plantlets and then plants.

- (g) Explain how plantlets can be formed from calli.

[2]

- (h) State **one** risk to the environment of growing genetically engineered crops and **one** possible risk to human health from eating these crops.

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[2]

[Total: 14]

[Turn over

- 3 Gene therapy is used to treat the genetic disorder consisting of ADA deficiency. Affected individuals are unable to produce the enzyme adenosine deaminase (ADA). Without this enzyme, T lymphocytes cannot provide immunity to infection.

Fig. 3.1 shows the processes involved in the treatment of ADA deficiency by gene therapy.

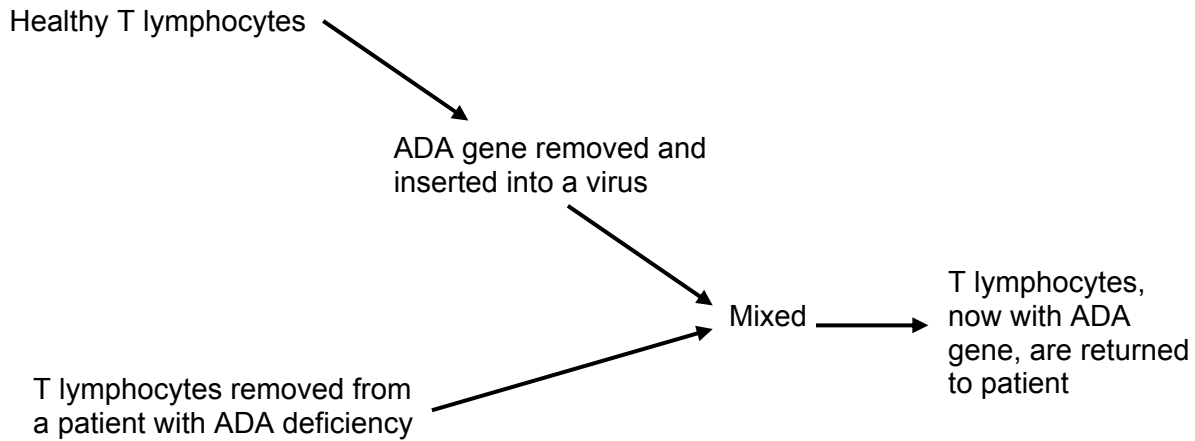


Fig. 3.1

- (a) Explain what is meant by “gene therapy”.

[2]

- (b) The ADA gene is inserted into a virus. Give **two** advantages of using viral method versus a non-viral method in gene therapy.

[2]

- (c) (i) Explain why the gene therapy method shown in Fig. 3.1 must be repeated at regular intervals for the treatment of ADA deficiency.

[2]

- (ii) Suggest a possible improvement to the process and explain how it will reduce the treatment intervals.

[2]

- (d) Individuals who have been treated by this method of gene therapy do not pass the normal functional ADA allele to their children. Explain why.

[1]

- (e) Discuss the social and ethical concerns regarding the use of gene therapy.

[3]

[Total: 12]

4 Planning Question

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Enzymes catalysing essentially irreversible reactions are potential sites of control in cellular respiration. One of these enzymes is phosphofructokinase, which can be regulated by the reversible binding of citrate to its allosteric site. (Citrate is produced as an intermediate compound during Krebs cycle.)

Using this information and your own knowledge, design an experiment to determine the effect of citrate on the rate of cellular respiration.

You must use:

- 10 mM citrate,
- purified homogenate of enzymes found in the cytosol,
- 5% glucose solution,
- distilled water,
- benedict's solution,
- apparatus shown in Fig. 4.1, can be used to separate proteins from ions and disaccharides.

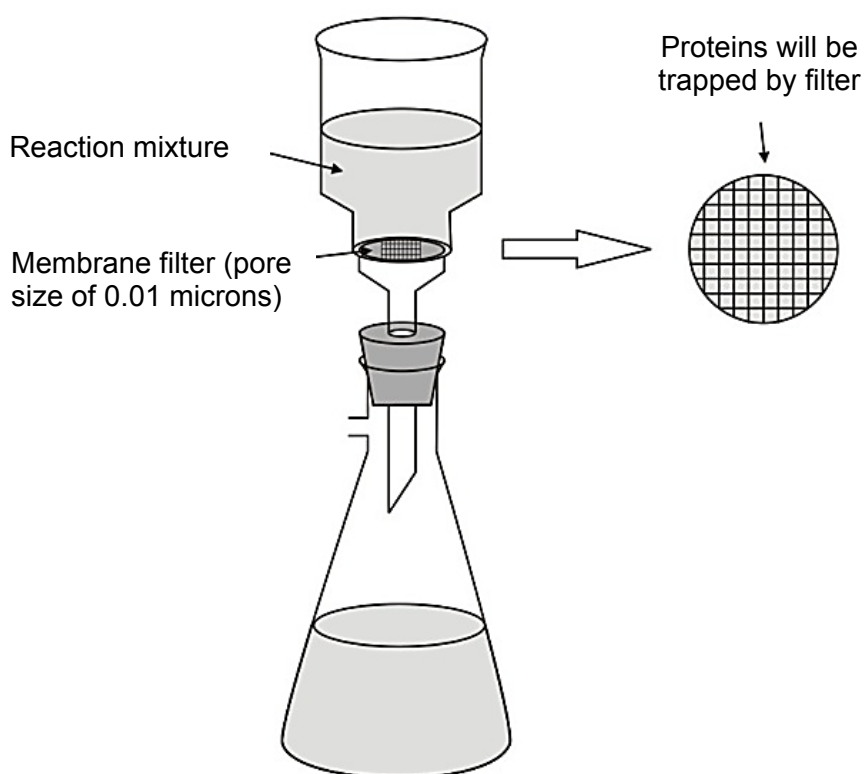


Fig. 4.1

[Turn over

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.,
- syringes,
- pipette fillers,
- white card,
- timer e.g. stopwatch or stop clock,
- thermometer,
- bunsen burner with tripod, gauze and bench mat.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment.

[Total: 12]

For
Examiner's
Use

[Turn over

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

5 Free-response question

Write your answers to this question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams where appropriate,
- must be in continuous prose, where appropriate,
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- (a)** Explain the advantages and disadvantages of the Polymerase Chain Reaction. [6]
- (b)** Explain how gel electrophoresis is able to show that a particular short tandem repeat (STR) has different lengths in different people. [6]
- (c)** Discuss the social and ethical issues of the Human Genome Project. [8]

[Total: 20]

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